Competing Risks: Prepayment vs. Default

Part E(i) — Mortgage Survival Analysis

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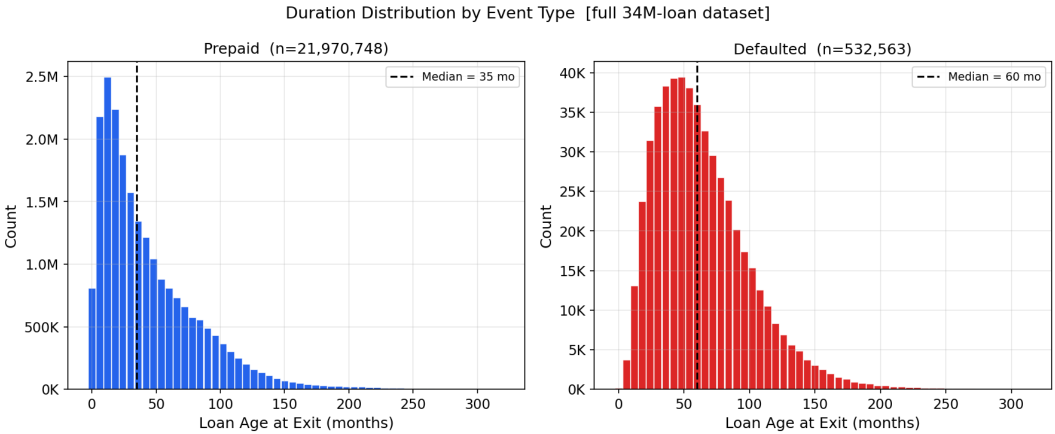
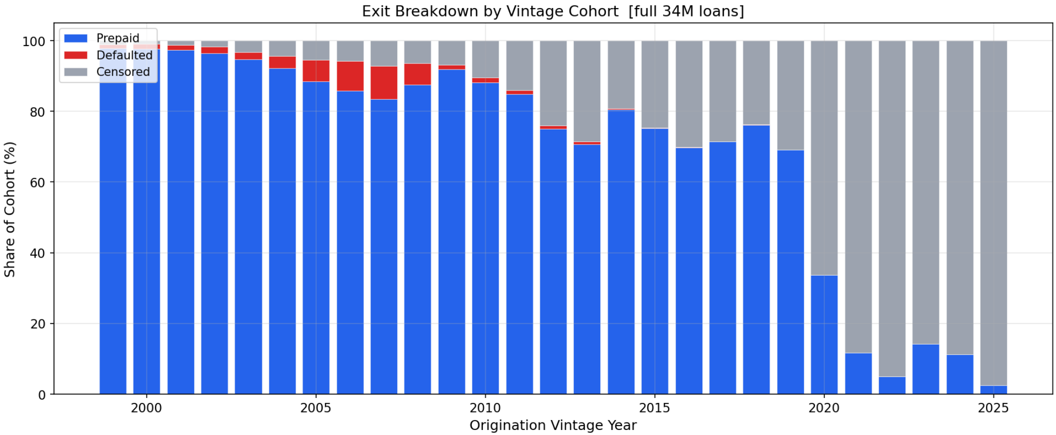
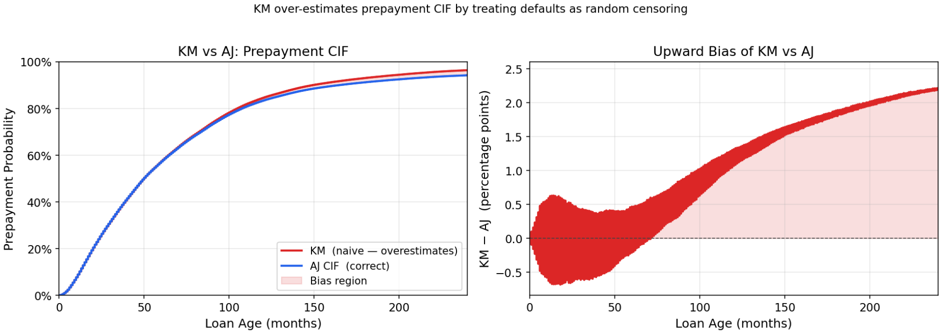
Two competing fates, two different borrower stories.



Vintage era — not FICO or LTV — is the dominant risk driver.

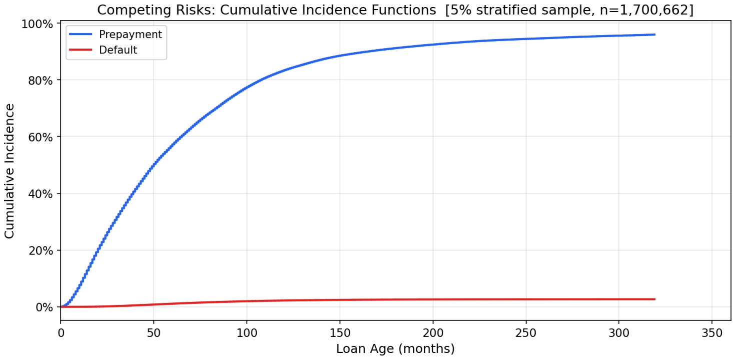


KM overstates prepayment probability because it ignores competing defaults.



Horizon	$\hat{F}_{\mathrm{KM}}$	$\hat{F}_{\mathrm{AJ}}$	Bias
5 yr	61.9%	61.4%	+0.45 pp
10 yr	84.6%	83.3%	+1.31 pp
20 yr	93.4%	91.8%	+2.23 pp

The Aalen–Johansen estimator gives the only coherent probability partition.



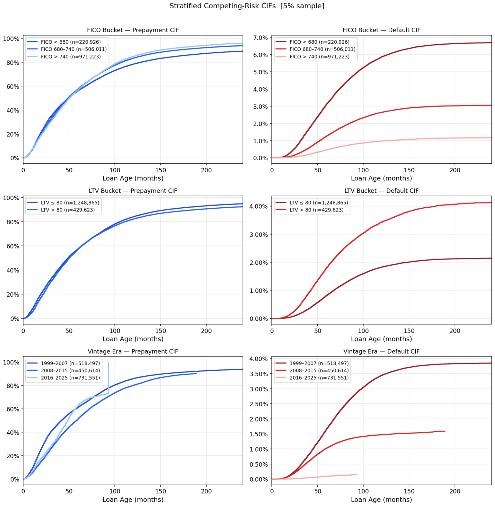
$$\hat{F}_k(t) = \sum_{t_j \le t} \hat{S}(t_j^-) \cdot \frac{d_{kj}}{n_j}$$

$$\sum_k \hat{F}_k + \hat{S} = 1$$

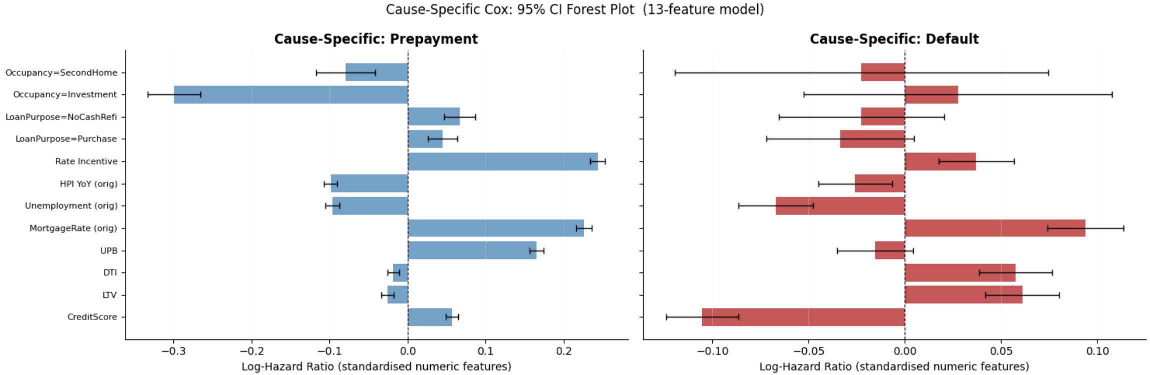
	Pre	Def
5 yr	61.4%	1.1%
10 yr	83.3%	2.2%
20 yr	91.8%	3.1%

37:1 ratio at 10 yr.
Burnout after month 40.

FICO separates default 6×; LTV separates default 2× — neither separates prepayment.



Prepayment and default respond to opposite covariate effects.

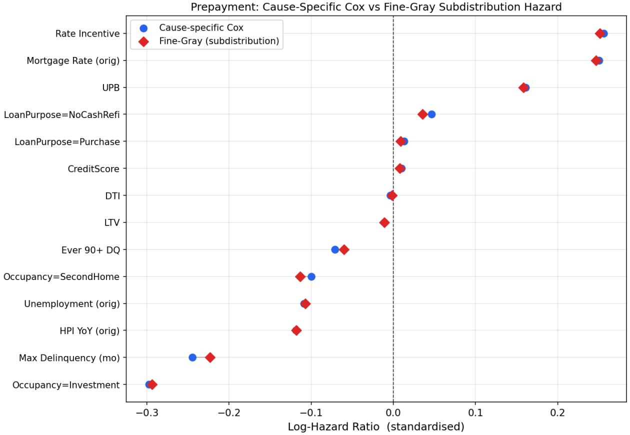


Four covariates flip sign — a combined model cancels both signals.

Covariate	Prepay $\hat{\beta}$	Default $\hat{\beta}$	
Rate incentive	$+0.256$	$+0.021$	(pure prepay signal)
FICO	$+0.010$	-0.047	**opposite**
LTV	-0.011	$+0.041$	**opposite**
Max Delinquency	-0.245	$+0.298$	**opposite**
Investment prop.	-0.297	$+0.270$	**opposite**

$e^{0.256} \approx 1.29$: a 1-SD rise in rate spread raises prepayment hazard 29%. $e^{0.298} \approx 1.35$: a 1-SD rise in max delinquency raises default hazard 35%.

Fine–Gray confirms cause-specific Cox; they differ only in high-default pools.



Max coefficient difference: $|\Delta| = 0.004$. When $F_{\text{def}}(t) \approx 0$: $\tilde{h}_1(t) \approx h_1(t)$.

Key Takeaways

1. **Competing risks are not optional**: 4 covariates flip sign between causes.
2. **KM bias reaches $+2.2$ pp at 20 years**: coherent forecasting requires Aalen–Johansen.
3. **Vintage dominates**: 2007 default CIF is $6\times$ long-run average; post-2020 prepayment is frozen.
4. **FICO separates default $6\times$; it barely separates prepayment**.
5. **Rate incentive is the dominant prepayment signal with zero default content**: $e^{0.256} \approx 1.29$.
6. **Fine–Gray $\approx$ cause-specific Cox here**; the distinction is material in subprime or crisis pools.

Thank you

Questions?

Fine & Gray (1999) *JASA* · Aalen & Johansen (1978) *Scand. J. Stat.* · Sadhwani et al. (2021) *J. Fin. Econometrics*